Build and tune the below 4 models to predict the label of an image from a given image on the yelp Photos dataset

https://business.yelp.com/data/resources/open-dataset/

Model Evaluation - 65%

kNN Model with Confusion Matrix and AUC - 15%

SVM Model with Confusion Matrix and AUC - 15% (can skip tuning on SVMs)

DNN Model with Confusion Matrix and AUC - 15%

CNN Model with Confusion Matrix and AUC - 20%

Data Preprocessing - 15%

Model Interpretability and Tuning - 20%

Run your jupyter notebook end to end and convert it to pdf before submitting

Also please add your models to the group GIT repository

Grading criteria

kNN

SVM

DNN

CNN

Rubric

Model Tuning	Identify the best feature set by comparing class separability ***10 points***	Lime used to study local interpretability ***14 points***	Analysis of model performance and areas for improvement identified ***17 points***	3 rounds of empirical tuning completed with well documented iterations ***20 points***
Data Preprocessing	Basic Preprocessing such as Scaling and Image Resizing Done ***6 points***	Further preprocessing done such as image augmentation ***9 points***	Advanced Preprocessing done to combine data from both images and other features ***12 points***	Detailed Preprocessing including Intensity Thresholds, Histogram equalization and Gaussian Blur ***15 points***
CNN	CNN implemented ***8 points***	CNN implemented well, AUC and Confusion matrix built ***12 points***	CNN implemented and tuned well, handling overfitting/underfitting, using both images if needed ***16 points***	Really-well tuned CNN, apt architecture to achieve strong performance, cross validation performed ***20 points***
DNN	Simple DNN built ***6 points***	DNN with Dropouts, multiple layers producing decent accuracy and recall, AUC and Confusion Matrix built ***9 points***	DNN tuned, using both images if needed, handling overfitting and underfitting ***12 points***	DNN with strong accuracy and recall, really well tuned, apt architecture for the dataset, cross validation performed ***15 points***
MECE	Not Present ***-25 points***	Not Clear ***-10 points***	All good ***0 points***	
Contribution to the group output	Very Insignificant Contribution ***-35 points***	Insignificant Contribution ***-15 points***	Significant Contribution ***0 points***	
Code Quality	No major errors, code	No warnings and code is	Aptly named variables, no very large outputs in the code making it	Code well commented, modularized

	takes time but can be read *-15 points*	easily readable *-10 points*	easy to follow, some code comments present *-5 points*	and well structured. Print outputs easily understandable *0 points*
SVM	SVM implemented *6 points*	SVM implemented well, AUC and Confusion matrix built *9 points*	SVM implemented and tuned well, handling overfitting/underfitting, using both images and other features if needed *12 points*	Really-well tuned SVM, apt data, features and parameters to achieve strong performance *15 points*
kNN	kNN implemented *6 points*	kNN implemented well, AUC and Confusion matrix built *9 points*	kNN implemented and tuned well, handling overfitting/underfitting, using both images and other features if needed *12 points*	Really-well tuned kNN, apt data, features and parameters to achieve strong performance *15 points*